CS604 · TEAM 06

# How our work answers the instructor's email

Slides: `cs604_team06_slides_v2.pptx` (20 slides) Report: `report/main.pdf` 6 November 2026

The instructor's email asks for three things:

1. Keep the backbone that the project brief recommends, unless there is a strong reason to change it.
2. Put most of the effort into adaptation methods: try them, compare them and improve them.
3. The grade does not depend only on the leaderboard. It also looks at how we studied the problem, how we explained our results, how we ran our experiments step by step, how we found and tested new ideas, what is new in our work, and what we learned, also from the ideas that failed.

The table shows where our slides and our report cover each point.

| What the instructor looks for | Where in the slides | Where in the report |
|---|---|---|
| **Keep the recommended backbone** | **Slide 2:** every experiment uses ViT-B/16, first trained on ImageNet-21k and then on ImageNet-1k. This is the model the brief recommends.<br>**Slide 4** (row "Nov 5"): we tested two versions trained only on ImageNet-21k. Their linear probes were weaker: 61.05% and 51.07%, against 61.94% for ours. | §3.1 |
| **Focus on adaptation methods** | **Slides 3 and 5:** we compare seven methods with the same settings.<br>**Slides 10 and 11:** our two final models in detail.<br>**Slides 16 to 20** (backup): the other five methods.<br>We also improved LoRA step by step: LoRA on all four linear layers ($+0.88$ points), half the learning rate ($-0.12$), 6 epochs instead of 3 ($-0.46$). | §3.2, §4.5, §4.7, §4.8 |
| **How we studied the problem** | **Slide 2:** four limits of the task: 10,000 classes, 45 photos per class, 1.2 GPU-hours per run, and a fixed backbone. Each one has a "So:" line that says what it means for our design. The photos on the slide show how different the photos of one class can be. | §1: the new head is large but has little data per class, and the budget is tight. This leads to three questions. |
| **How we explained our results** | **Slide 5:** where the trained weights sit matters more than how many there are. Training memory depends on how far back the gradients must go, not on how many weights we train.<br>**Slide 8:** lower training loss but lower accuracy, so the model memorizes the training photos.<br>**Slide 9:** one reason for each change in our recipe.<br>**Slide 13:** why bigger images help: random crops make objects look bigger during training, so bigger test images match them better (FixRes). | §4.5 to §4.9 |
| **How we ran experiments step by step** | **Slide 4:** eight rounds. Each round asks one question, and we mark whether the idea worked or failed. | §4.2 and Table 2 |
| **How we found and tested new ideas** | **Slides 6 to 8:** LoRA's training loss went up $\rightarrow$ we halved the learning rate, but nothing changed ($-0.12$) $\rightarrow$ so we tested the head $\rightarrow$ the prototype head gave $+2.5$ $\rightarrow$ a random head served as a control.<br>**Slide 13:** first we only tested with bigger images ($+2.1$), then we also trained with bigger images (288 pixels, $+4.4$). | §4.2, §4.6, §4.9.<br>§4.8: our idea for why LoRA gets worse with longer training (the head memorizes), and the run that will test it (notebook 10). |

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|---|---|---|
| **What is new in our work** | **Slide 14, point 1:** the usual LP-FT start hurt all three methods we tested, while a head built from class prototypes added 2.1 to 3.1 points.<br>**Slides 6 and 7** name the papers behind both ideas (LP-FT: Kumar et al., ICLR 2022; imprinting: Qi et al., CVPR 2018). So what we claim as new is the finding, not the method.<br>**Slide 14, point 2:** where you train matters more than how much you train. | §1 (our contributions) and §4.6 |
| **Failed ideas and lessons learned** | **Slides 4 and 9:** the ideas that failed (half the learning rate: $-0.12$; 6 LoRA epochs: $-0.46$).<br>**Slide 14:** point 3, failed ideas were useful. Point 4, our limits: we ran each setting once, differences under about 0.3 points are noise, and all efficient methods used one learning rate that we did not tune. Point 5, the lesson we found after the deadline, and our next step. | §4.2 and §5 (Lessons from the process; Limitations) |

**Also prepared.** The speaker notes of slide 15 have short answers to questions we expect. For example: why LoRA gets worse with longer training, and why a random head did not hurt us, although the LP-FT paper warns against it.